Column Space, Null Space and Rank-Nullity

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Column space and null space

Definition 0.1 (Column space (range, image))**.** The **column space** [1] of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{C}(\mathbf{A}) \stackrel{\text{def}}{=} \{\mathbf{A}\underset{\sim}{x} : \underset{\sim}{x} \in \mathbb{R}^n\},$$

the set of all vectors $\mathbf{A}\underset{\sim}{x}$ in $\mathbb{R}^m$. It is also called the **range** or **image** of $\mathbf{A}$.

Definition 0.2 (Row space)**.** The **row space** of an $m \times n$ matrix $\mathbf{A}$ is $\mathcal{C}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^n$.

Example 0.1 (The column space of a rank-one matrix)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix}.$$

For any $\underset{\sim}{x} \in \mathbb{R}^3$,

$$\begin{aligned} \mathbf{A}\underset{\sim}{x} &= (x_1 - 2x_2 - 2x_3,\ 3\,(x_1 - 2x_2 - 2x_3)) \\ &= (x_1 - 2x_2 - 2x_3)\,(1, 3) \end{aligned}$$

(Definition 13 in Matrices). The number $x_1 - 2x_2 - 2x_3$ can be any real number (take $x_2 = 0$ and $x_3 = 0$), so $\mathcal{C}(\mathbf{A}) = \{c\,(1, 3) : c \in \mathbb{R}\}$, a line in $\mathbb{R}^2$. The vector $(1, 0)$ is not in $\mathcal{C}(\mathbf{A})$: $c\,(1, 3)$ has second entry $3c$, which is 0 only if $c = 0$, and then the first entry is 0, not 1.

Theorem 0.1 (The column space is the span of the columns)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix with columns* $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$*, then*

$$\mathcal{C}(\mathbf{A}) = \operatorname{span}\{\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n\},$$

so $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$, and the row space $\mathcal{C}(\mathbf{A}^\top)$ is a subspace of $\mathbb{R}^n$.

Proof

Proof. By Theorem 3 in Subspaces and Rank, $\mathbf{A}\underset{\sim}{x} = x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n$, and as $\underset{\sim}{x}$ ranges over $\mathbb{R}^n$, the coefficients $x_1, \ldots, x_n$ range over all lists of n numbers. So the set of all $\mathbf{A}\underset{\sim}{x}$ (Definition 0.1) is the set of all linear combinations of the columns, which is their span (Definition 5 in Subspaces and Rank). A span is a subspace (Theorem 2 in Subspaces and Rank). Applying the same argument to $\mathbf{A}^\top$, whose columns are the rows of $\mathbf{A}$, shows the row space is a subspace of $\mathbb{R}^n$. □

Example 0.2 (Reading off the column space and row space from the columns and rows)**.** For $\mathbf{A}$ in Example 0.1, the columns are $(1, 3)$, $(-2, -6) = -2\,(1, 3)$ and $(-2, -6) = -2\,(1, 3)$, so by Theorem 0.1 $\mathcal{C}(\mathbf{A}) = \operatorname{span}\{(1, 3)\}$, the same line found in Example 0.1. The rows are $(1, -2, -2)$ and $(3, -6, -6) = 3\,(1, -2, -2)$, so the row space is $\operatorname{span}\{(1, -2, -2)\}$, a line in $\mathbb{R}^3$.

Definition 0.3 (Null space (kernel))**.** The **null space** of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{N}(\mathbf{A}) \stackrel{\text{def}}{=} \{\underset{\sim}{x} \in \mathbb{R}^n : \mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m\},$$

the set of solutions of the linear system $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$. It is also called the **kernel** of $\mathbf{A}$.

Definition 0.4 (Left null space)**.** The **left null space** of an $m \times n$ matrix $\mathbf{A}$ is $\mathcal{N}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^m$.

Example 0.3 (The null space and left null space of a rank-one matrix)**.** For $\mathbf{A}$ in Example 0.1, $\mathbf{A}\underset{\sim}{x} = (x_1 - 2x_2 - 2x_3,\ 3\,(x_1 - 2x_2 - 2x_3))$, so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_2$ exactly when $x_1 = 2x_2 + 2x_3$. Writing

$$\begin{aligned}\underset{\sim}{x} &= (2x_2 + 2x_3, x_2, x_3)\\ &= x_2\,(2, 1, 0) + x_3\,(2, 0, 1)\end{aligned}$$

shows $\mathcal{N}(\mathbf{A}) = \operatorname{span}\{(2, 1, 0), (2, 0, 1)\}$, a plane in $\mathbb{R}^3$.
For the left null space, $\mathbf{A}^\top y = (y_1 + 3y_2,\ -2\,(y_1 + 3y_2),\ -2\,(y_1 + 3y_2))$, which is $\underset{\sim}{0}_3$ exactly when $y_1 = -3y_2$, so $\mathcal{N}(\mathbf{A}^\top) = \operatorname{span}\{(-3, 1)\}$, a line in $\mathbb{R}^2$.

Theorem 0.2 (A null space is a subspace)**.** *For any $m \times n$ matrix $\mathbf{A}$, $\mathcal{N}(\mathbf{A})$ (Definition 0.3) is a subspace of $\mathbb{R}^n$ (Definition 4 in Subspaces and Rank).*

Proof

Proof. $\mathbf{A}\underset{\sim}{0}_n = \underset{\sim}{0}_m$, so $\underset{\sim}{0}_n \in \mathcal{N}(\mathbf{A})$, and the null space is not empty. If $\underset{\sim}{u}, \underset{\sim}{w} \in \mathcal{N}(\mathbf{A})$ and $c \in \mathbb{R}$, then

$$\begin{aligned} \mathbf{A}\left(\underset{\sim}{u} + \underset{\sim}{w}\right) &= \mathbf{A}\underset{\sim}{u} + \mathbf{A}\underset{\sim}{w} && \text{(Theorem 6 in Matrices)} \\ &= \underset{\sim}{0}_m + \underset{\sim}{0}_m && (\underset{\sim}{u} \text{ and } \underset{\sim}{w} \text{ are in the null space}) \\ &= \underset{\sim}{0}_m && (\text{adding } \underset{\sim}{0}_m \text{ changes nothing}) \end{aligned}$$

and, for each entry $i = 1, \ldots, m$,

$$\begin{aligned} \left(\mathbf{A}\left(c\underset{\sim}{u}\right)\right)_i &= a_{i1}(cu_1) + \cdots + a_{in}(cu_n) && \text{(Definition 13 in Matrices)} \\ &= c\left(a_{i1}u_1 + \cdots + a_{in}u_n\right) && (\text{factor } c \text{ out of each term}) \\ &= c\left(\mathbf{A}\underset{\sim}{u}\right)_i && \text{(Definition 13 in Matrices)} \\ &= c \cdot 0 && (\underset{\sim}{u} \text{ is in the null space}) \\ &= 0, && (\text{arithmetic}) \end{aligned}$$

so $\mathbf{A}\left(c\underset{\sim}{u}\right) = \underset{\sim}{0}_m$. Therefore $\underset{\sim}{u} + \underset{\sim}{w}$ and $c\underset{\sim}{u}$ are in $\mathcal{N}(\mathbf{A})$. □

Example 0.4 (The solutions of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ with $\underset{\sim}{b} \neq \underset{\sim}{0}$ are not a subspace)**.** In Example 0.3, $(2, 1, 0)$ and $(2, 0, 1)$ are in $\mathcal{N}(\mathbf{A})$, and so is their sum $(4, 1, 1)$, since $4 - 2 \cdot 1 - 2 \cdot 1 = 0$. By contrast, the solutions of $\mathbf{A}\underset{\sim}{x} = (1, 3)$ for the same $\mathbf{A}$ are the vectors with $x_1 - 2x_2 - 2x_3 = 1$. That set does not contain $\underset{\sim}{0}_3$, so by Theorem 1 in Subspaces and Rank it is not a subspace.

Theorem 0.3 (Independent columns exactly when the null space is trivial)**.** *For an $m \times n$ matrix $\mathbf{A}$, the following are equivalent:*

1. *The columns of $\mathbf{A}$ are linearly independent (Definition 1 in Subspaces and Rank).*
2. *$\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ only for $\underset{\sim}{x} = \underset{\sim}{0}_n$.*
3. *$\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}$ (Definition 0.3).*

So $\mathbf{A}$ has full column rank (Definition 3 in Subspaces and Rank) exactly when its null space contains only the zero vector.

Proof

Proof. Write $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_n$ for the columns of $\mathbf{A}$. For $\underset{\sim}{x} = (x_1, \dots, x_n)$, the matrix-vector product is the combination of the columns with weights $x_1, \dots, x_n$:

$$\mathbf{A}\underset{\sim}{x} = x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n.$$

So $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ says exactly that $x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n = \underset{\sim}{0}_m$. By Definition 1 in Subspaces and Rank, the columns are linearly independent exactly when the only weights with this property are $x_1 = \cdots = x_n = 0$. That is statement 2, so statements 1 and 2 are equivalent. Statement 2 says that the only vector in $\mathcal{N}(\mathbf{A})$ is $\underset{\sim}{0}_n$, which is statement 3. The last sentence is the definition of full column rank (Definition 3 in Subspaces and Rank). □

Example 0.5 (Checking independence through the null space)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$,

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} x_1 + x_2 \\ x_1 + 2x_2 \\ x_1 + 3x_2 \end{bmatrix}.$$

This is $\underset{\sim}{0}_3$ only if $x_2 = 0$ (subtract the first entry from the second), and then $x_1 = 0$. So $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_2\}$ and the two columns are linearly independent.
For $\mathbf{B} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$, $\mathbf{B}(2, -1) = \underset{\sim}{0}_3$, so $\mathcal{N}(\mathbf{B})$ contains a nonzero vector and the columns of $\mathbf{B}$ are linearly dependent.

Theorem 0.4 (The row space and null space share only the zero vector)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{C}(\mathbf{A}^\top) \cap \mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}.$$

Proof

Proof. Both sets are subspaces, by Theorem 0.1 and Theorem 0.2, so both contain $\underset{\sim}{0}_n$ (Theorem 1 in Subspaces and Rank).
Now take any $\underset{\sim}{x}$ in both sets. Being in the row space means $\underset{\sim}{x} = \mathbf{A}^\top\underset{\sim}{u}$ for some $\underset{\sim}{u} \in \mathbb{R}^m$ (Definition 0.1), and being in the null space means $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ (Definition 0.3). Then

$$
\begin{aligned}
\underset{\sim}{x}^\top \underset{\sim}{x} &= \left(\mathbf{A}^\top \underset{\sim}{u}\right)^\top \underset{\sim}{x} && \text{(substitute } \underset{\sim}{x} = \mathbf{A}^\top \underset{\sim}{u} \text{ in the first factor)} \\
&= \left(\underset{\sim}{u}^\top \left(\mathbf{A}^\top\right)^\top\right) \underset{\sim}{x} && \text{(Theorem 17 in Matrices)} \\
&= \left(\underset{\sim}{u}^\top \mathbf{A}\right) \underset{\sim}{x} && \text{(transposing twice returns the original matrix)} \\
&= \underset{\sim}{u}^\top \left(\mathbf{A}\underset{\sim}{x}\right) && \text{(Theorem 5 in Matrices)} \\
&= \underset{\sim}{u}^\top \underset{\sim}{0}_m && \text{(substitute } \mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m\text{)} \\
&= 0. && \text{(every term of the inner product is 0)}
\end{aligned}
$$

But $\underset{\sim}{x}^\top \underset{\sim}{x} = x_1^2 + \cdots + x_n^2$, which is 0 only when every $x_i = 0$. So $\underset{\sim}{x} = \underset{\sim}{0}_n$. □

Example 0.6 (The row space and null space in Example 0.3)**.** For $\mathbf{A}$ in Example 0.1, the row space is $\text{span}\{(1, -2, -2)\}$ and the null space is $\text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.3). A vector $c\,(1, -2, -2)$ of the row space is in the null space only if

$$
\begin{aligned}
\mathbf{A}\,(c\,(1, -2, -2)) &= c\,(1 - 2\cdot(-2) - 2\cdot(-2),\ 3\,(1 - 2\cdot(-2) - 2\cdot(-2))) && \text{(Example 0.3's formula for } \mathbf{A}\underset{\sim}{x}\text{)} \\
&= c\,(9, 27) && \text{(arithmetic)}
\end{aligned}
$$

is $\underset{\sim}{0}_2$, that is, only if $c = 0$. So the two subspaces share only $\underset{\sim}{0}_3$.

Definition 0.5 (Cross-product matrix)**.** Let $\mathbf{A}$ be an $m \times n$ matrix with columns $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ and $\mathbf{B}$ an $m \times q$ matrix with columns $\underset{\sim}{b}_1, \ldots, \underset{\sim}{b}_q$. The **cross-product matrix** of $\mathbf{A}$ and $\mathbf{B}$ is the $n \times q$ matrix $\mathbf{A}^\top\mathbf{B}$, whose entry (i, j) is the dot product $\underset{\sim}{a}_i \cdot \underset{\sim}{b}_j$.

This is what R's `crossprod(A, B)` computes, and `crossprod(A)` is $\mathbf{A}^\top\mathbf{A}$. In NumPy it has no name of its own: write `A.T @ B`.

For two column vectors it is the 1×1 matrix holding their dot product. It is unrelated to the cross product $\underset{\sim}{u} \times \underset{\sim}{v}$ of two vectors in $\mathbb{R}^3$, and differs from the Frobenius inner product $\text{tr}(\mathbf{A}^\top\mathbf{B})$, which sums the diagonal of this matrix into a single number.

Example 0.7 (A cross-product matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Then $\mathbf{A}^\top\mathbf{B}$ is 2×1, and

$$\begin{aligned}
\mathbf{A}^\top \mathbf{B} &= \begin{bmatrix} \underset{\sim}{a}_1 \cdot \underset{\sim}{b}_1 \\ \underset{\sim}{a}_2 \cdot \underset{\sim}{b}_1 \end{bmatrix} && \text{(Definition 0.5)} \\
&= \begin{bmatrix} 1+2+0 \\ 0+2+3 \end{bmatrix} && \text{(multiply and add)} \\
&= \begin{bmatrix} 3 \\ 5 \end{bmatrix}. && \text{(add)}
\end{aligned}$$

Definition 0.6 (Gram matrix)**.** The **Gram matrix** [2] of an $m \times n$ matrix $\mathbf{A}$ is the $n \times n$ matrix $\mathbf{A}^\top \mathbf{A}$, the cross-product matrix of $\mathbf{A}$ with itself.

Example 0.8 (A Gram matrix and its entries)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, with columns $\underset{\sim}{a}_1 = (1,1,0)$ and $\underset{\sim}{a}_2 = (0,1,1)$. Then

$$\begin{aligned}
\mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} && \text{(Definition 3 in Matrices)} \\
&= \begin{bmatrix} 1\cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 \end{bmatrix} && \text{(Definition 7 in Matrices)} \\
&= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. && \text{(multiply and add)}
\end{aligned}$$

Row i of $\mathbf{A}^\top$ is column i of $\mathbf{A}$, so entry (i,j) of the Gram matrix is the dot product $\underset{\sim}{a}_i \cdot \underset{\sim}{a}_j$: for instance,

$$\begin{aligned}
\underset{\sim}{a}_1 \cdot \underset{\sim}{a}_2 &= 0 + 1 + 0 \\
&= 1
\end{aligned}$$

is entry $(1,2)$.

Theorem 0.5 ($\mathbf{A}^\top \mathbf{A}$ has the same null space as $\mathbf{A}$)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}).$$

Proof

Proof. $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}^\top \mathbf{A})$**.** If $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$, then

$$\begin{aligned}\mathbf{A}^\top \mathbf{A}\underset{\sim}{x} &= \mathbf{A}^\top \underset{\sim}{0}_m \\ &= \underset{\sim}{0}_n.\end{aligned}$$

$\mathcal{N}(\mathbf{A}^\top\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$. If $\mathbf{A}^\top\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_n$, then

$$\begin{aligned}
0 &= \underset{\sim}{x}^\top \underset{\sim}{0}_n && \text{(every term of the inner product is 0)} \\
&= \underset{\sim}{x}^\top (\mathbf{A}^\top\mathbf{A}\underset{\sim}{x}) && \text{(substitute } \underset{\sim}{0}_n = \mathbf{A}^\top\mathbf{A}\underset{\sim}{x}\text{)} \\
&= (\underset{\sim}{x}^\top\mathbf{A}^\top)(\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{A}\underset{\sim}{x})^\top (\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 17 in Matrices)} \\
&= (\mathbf{A}\underset{\sim}{x}) \cdot (\mathbf{A}\underset{\sim}{x}) && \text{(Example 6 in Matrices)} \\
&= \|\mathbf{A}\underset{\sim}{x}\|^2. && \text{(square both sides of Equation 2 in Vectors)}
\end{aligned}$$

A vector whose Euclidean norm is 0 has every entry 0, so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$. □

Example 0.9 (The null space of a Gram matrix)**.** For $\mathbf{A}$ in Example 0.1,

$$\begin{aligned}\mathbf{A}^\top\mathbf{A} &= \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix} \\ &= \begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix}.\end{aligned}$$

Every row is a multiple of $(1, -2, -2)$, so $\mathbf{A}^\top\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_3$ exactly when $x_1 - 2x_2 - 2x_3 = 0$, the same condition as for $\mathcal{N}(\mathbf{A})$ in Example 0.3.

Rank-nullity and rank factorization

i Source

This section is adapted from Zhou [3], used under the MIT License (see the license text in Section 2 in Subspaces and Rank).

Theorem 0.6 (The rank is the dimension of the column space)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\operatorname{rank}(\mathbf{A}) = \dim(\mathcal{C}(\mathbf{A})).$$

Proof

Proof. Let $r = \text{rank}(\mathbf{A})$ (Definition 2 in Subspaces and Rank), and choose r linearly independent columns of $\mathbf{A}$; by Definition 2 in Subspaces and Rank, no larger set of columns is linearly independent. We show the chosen columns are a basis of $\mathcal{C}(\mathbf{A})$ (Definition 6 in Subspaces and Rank).

Every column is in the span of the chosen columns. A chosen column is in that span trivially. If some unchosen column $\underset{\sim}{a}_j$ were not in the span, then adding it to the chosen columns would give $r+1$ linearly independent columns (Theorem 8 in Subspaces and Rank), more than r. So every column of $\mathbf{A}$ is in the span of the chosen columns.

The chosen columns span $\mathcal{C}(\mathbf{A})$. The span of the chosen columns is a subspace (Theorem 2 in Subspaces and Rank), so it is closed under addition and scalar multiplication, and it contains every linear combination of the columns of $\mathbf{A}$. By Theorem 0.1, those combinations make up $\mathcal{C}(\mathbf{A})$, so $\mathcal{C}(\mathbf{A})$ is contained in the span of the chosen columns. The chosen columns are themselves columns of $\mathbf{A}$, so their span is contained in $\mathcal{C}(\mathbf{A})$, and the two sets are equal.

The chosen columns are linearly independent and span $\mathcal{C}(\mathbf{A})$, so they are a basis of it, and $\dim(\mathcal{C}(\mathbf{A})) = r$ (Definition 9 in Subspaces and Rank). If $r = 0$, every column is $\underset{\sim}{0}_m$, $\mathcal{C}(\mathbf{A}) = \{\underset{\sim}{0}_m\}$, and both sides are 0. □

Example 0.10 (The rank of the matrix in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1,3)\}$ (Example 0.2), and $(1,3)$ is linearly independent because it is nonzero, so it is a basis and $\mathcal{C}(\mathbf{A})$ has dimension 1. So $\text{rank}(\mathbf{A}) = 1$, by Theorem 0.6. Directly: the first column $(1,3)$ is linearly independent on its own, and any two columns are dependent, because each column is a multiple of $(1,3)$.

Definition 0.7 (Nullity)**.** The **nullity** [4] of a matrix $\mathbf{A}$ is the dimension of its null space (Definition 0.3):

$$\text{nullity}(\mathbf{A}) \stackrel{\text{def}}{=} \dim(\mathcal{N}(\mathbf{A})).$$

Example 0.11 (The nullity of the matrix in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $\mathcal{N}(\mathbf{A}) = \text{span}\{(2,1,0),(2,0,1)\}$ (Example 0.3). Those two vectors are linearly independent: $c_1(2,1,0) + c_2(2,0,1) = (2c_1 + 2c_2, c_1, c_2)$, which is $\underset{\sim}{0}_3$ only if

$$\begin{aligned} c_1 &= c_2 \\ &= 0. \end{aligned}$$

So they are a basis of $\mathcal{N}(\mathbf{A})$, and $\text{nullity}(\mathbf{A}) = 2$. The 2×2 identity matrix has nullity 0, because $\mathbf{I}\underset{\sim}{x} = \underset{\sim}{x}$ is $\underset{\sim}{0}_2$ only when $\underset{\sim}{x} = \underset{\sim}{0}_2$.

Theorem 0.7 (Rank-nullity theorem)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\operatorname{rank}(\mathbf{A}) + \operatorname{nullity}(\mathbf{A}) = n.$$

Proof

Proof. Let $\nu = \operatorname{nullity}(\mathbf{A})$. Choose a basis $\underset{\sim}{x}_1, \dots, \underset{\sim}{x}_\nu$ of $\mathcal{N}(\mathbf{A})$, and extend it to a basis $\underset{\sim}{x}_1, \dots, \underset{\sim}{x}_\nu, \underset{\sim}{y}_1, \dots, \underset{\sim}{y}_{n-\nu}$ of $\mathbb{R}^n$ (Theorem 9 in Subspaces and Rank); it has n vectors, because every basis of $\mathbb{R}^n$ has $\dim(\mathbb{R}^n) = n$ vectors (Theorem 7 in Subspaces and Rank, Example 16 in Subspaces and Rank). We show that $\mathbf{A}\underset{\sim}{y}_1, \dots, \mathbf{A}\underset{\sim}{y}_{n-\nu}$ are a basis of $\mathcal{C}(\mathbf{A})$, using Theorem 4 in Subspaces and Rank to apply $\mathbf{A}$ to linear combinations.

They are linearly independent. Suppose $\sum_{i=1}^{n-\nu} v_i \, \mathbf{A}\underset{\sim}{y}_i = \underset{\sim}{0}_m$. Then $\mathbf{A}\left(\sum_i v_i \underset{\sim}{y}_i\right) = \underset{\sim}{0}_m$ (Theorem 4 in Subspaces and Rank), so $\sum_i v_i \underset{\sim}{y}_i \in \mathcal{N}(\mathbf{A})$, and it equals $\sum_{j=1}^{\nu} u_j \underset{\sim}{x}_j$ for some numbers u_j, since the $\underset{\sim}{x}_j$ span $\mathcal{N}(\mathbf{A})$. Moving everything to one side,

$$\sum_{j=1}^{\nu} (-u_j)\, \underset{\sim}{x}_j + \sum_{i=1}^{n-\nu} v_i \underset{\sim}{y}_i = \underset{\sim}{0}_n,$$

and because the $\underset{\sim}{x}$'s and $\underset{\sim}{y}$'s together form a basis of $\mathbb{R}^n$, which is linearly independent, every coefficient is 0; in particular every $v_i = 0$.

They span $\mathcal{C}(\mathbf{A})$. Take any $\underset{\sim}{w} \in \mathcal{C}(\mathbf{A})$, so $\underset{\sim}{w} = \mathbf{A}\underset{\sim}{z}$ for some $\underset{\sim}{z} \in \mathbb{R}^n$ (Definition 0.1). Write $\underset{\sim}{z} = \sum_j a_j \underset{\sim}{x}_j + \sum_i b_i \underset{\sim}{y}_i$ in the basis of $\mathbb{R}^n$. Then

$$\begin{aligned}
\underset{\sim}{w} &= \mathbf{A}\left(\sum_{j=1}^{\nu} a_j \underset{\sim}{x}_j + \sum_{i=1}^{n-\nu} b_i \underset{\sim}{y}_i\right) && \text{(substitute } \underset{\sim}{z}\text{)} \\
&= \sum_{j=1}^{\nu} a_j \, \mathbf{A}\underset{\sim}{x}_j + \sum_{i=1}^{n-\nu} b_i \, \mathbf{A}\underset{\sim}{y}_i && \text{(Theorem 4 in Subspaces and Rank)} \\
&= \sum_{j=1}^{\nu} a_j \, \underset{\sim}{0}_m + \sum_{i=1}^{n-\nu} b_i \, \mathbf{A}\underset{\sim}{y}_i && \text{(each } \underset{\sim}{x}_j \in \mathcal{N}(\mathbf{A})\text{)} \\
&= \sum_{i=1}^{n-\nu} b_i \, \mathbf{A}\underset{\sim}{y}_i, && \text{(drop the zero terms)}
\end{aligned}$$

a linear combination of $\mathbf{A}\underset{\sim}{y}_1, \dots, \mathbf{A}\underset{\sim}{y}_{n-\nu}$. These vectors are in $\mathcal{C}(\mathbf{A})$, so their span is exactly $\mathcal{C}(\mathbf{A})$.

So $\dim(\mathcal{C}(\mathbf{A})) = n - \nu$, and $\operatorname{rank}(\mathbf{A}) = n - \nu$ by Theorem 0.6. □

Example 0.12 (Checking rank-nullity on the matrix in Example 0.1)**.** The matrix $\mathbf{A}$ in Example 0.1 has $n = 3$ columns, rank 1 (Example 0.10) and nullity 2 (Example 0.11), and $1 + 2 = 3$. The 2×2 identity matrix has rank 2 and nullity 0 (Example 0.11), and $2 + 0 = 2$.

Theorem 0.8 (Multiplying on the right cannot increase the rank)**.** *If* $\mathbf{A}$ *is* $m \times n$ *and* $\mathbf{B}$ *is* $n \times k$*, then*

$$\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank}(\mathbf{A}).$$

Proof

Proof. $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$**.** Any vector in $\mathcal{C}(\mathbf{AB})$ is $(\mathbf{AB})\underset{\sim}{x}$ for some $\underset{\sim}{x} \in \mathbb{R}^k$ (Definition 0.1), and

$$(\mathbf{AB})\underset{\sim}{x} = \mathbf{A}(\mathbf{B}\underset{\sim}{x}) \quad \text{(Theorem 5 in Matrices)}$$

is $\mathbf{A}$ times the vector $\mathbf{B}\underset{\sim}{x} \in \mathbb{R}^n$, so it is in $\mathcal{C}(\mathbf{A})$.
Compare dimensions. Both column spaces are subspaces (Theorem 0.1), so

$$\begin{aligned} \operatorname{rank}(\mathbf{AB}) &= \dim(\mathcal{C}(\mathbf{AB})) && \text{(Theorem 0.6)} \\ &\leq \dim(\mathcal{C}(\mathbf{A})) && \text{(Theorem 10 in Subspaces and Rank, part 2)} \\ &= \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.6)} \end{aligned}$$

□

Example 0.13 (A product can lose rank)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{AB} = \begin{bmatrix} 1\cdot 0 + 0\cdot 1 & 1\cdot 0 + 0\cdot 0 \\ 0\cdot 0 + 0\cdot 1 & 0\cdot 0 + 0\cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\mathbf{A}$ has rank 1, but $\mathbf{AB}$ has rank 0, so the inequality in Theorem 0.8 can be strict. With the 2×2 identity in place of $\mathbf{B}$, $\mathbf{AI} = \mathbf{A}$ keeps rank 1, and the inequality is an equality.

Theorem 0.9 (Fundamental theorem of ranks)**.** *For any* $m \times n$ *matrix* $\mathbf{A}$*,*

$$\begin{aligned} \operatorname{rank}(\mathbf{A}) &= \operatorname{rank}(\mathbf{A}^\top) \\ &= \operatorname{rank}(\mathbf{A}^\top\mathbf{A}) \\ &= \operatorname{rank}(\mathbf{A}\mathbf{A}^\top). \end{aligned}$$

In particular, the largest number of linearly independent rows of $\mathbf{A}$ *equals the largest number of linearly independent columns.*

Proof

Proof. $\operatorname{rank}(\mathbf{A}^\top\mathbf{A}) = \operatorname{rank}(\mathbf{A})$**.** Both matrices have n columns, and $\mathcal{N}(\mathbf{A}^\top\mathbf{A}) = \mathcal{N}(\mathbf{A})$ (Theorem 0.5), so they have the same nullity (Definition 0.7). Then

$$\begin{aligned} \operatorname{rank}(\mathbf{A}^\top\mathbf{A}) &= n - \operatorname{nullity}(\mathbf{A}^\top\mathbf{A}) && \text{(Theorem 0.7)} \\ &= n - \operatorname{nullity}(\mathbf{A}) && \text{(equal null spaces)} \\ &= \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.7)} \end{aligned}$$

$\operatorname{rank}(\mathbf{A}\mathbf{A}^\top) = \operatorname{rank}(\mathbf{A}^\top)$**.** Apply the previous step to the $n \times m$ matrix $\mathbf{A}^\top$, using $(\mathbf{A}^\top)^\top = \mathbf{A}$.

Chain the inequalities.

$$\begin{aligned} \operatorname{rank}(\mathbf{A}) &= \operatorname{rank}(\mathbf{A}^\top\mathbf{A}) && \text{(first step)} \\ &\le \operatorname{rank}(\mathbf{A}^\top) && \text{(Theorem 0.8, with } \mathbf{A}^\top \text{ on the left)} \\ &= \operatorname{rank}(\mathbf{A}\mathbf{A}^\top) && \text{(second step)} \\ &\le \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.8, with } \mathbf{A} \text{ on the left)} \end{aligned}$$

The chain starts and ends at $\operatorname{rank}(\mathbf{A})$, so every quantity in it equals $\operatorname{rank}(\mathbf{A})$. The columns of $\mathbf{A}^\top$ are the rows of $\mathbf{A}$, which gives the statement about rows. □

Example 0.14 (Four equal ranks)**.** For $\mathbf{A}$ in Example 0.1, $\operatorname{rank}(\mathbf{A}) = 1$ (Example 0.10). Its transpose $\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}$ has second column 3 times its first, so its rank is 1. $\mathbf{A}^\top\mathbf{A}$ in Example 0.9 has every row a multiple of $(1, -2, -2)$, so its rank is 1 too, and

$$\begin{aligned} \mathbf{A}\mathbf{A}^\top &= \begin{bmatrix} 1\cdot 1 + (-2)(-2) + (-2)(-2) & 1\cdot 3 + (-2)(-6) + (-2)(-6) \\ 3\cdot 1 + (-6)(-2) + (-6)(-2) & 3\cdot 3 + (-6)(-6) + (-6)(-6) \end{bmatrix} \\ &= \begin{bmatrix} 9 & 27 \\ 27 & 81 \end{bmatrix}, \end{aligned}$$

whose second column is 3 times its first, so its rank is 1.

Corollary 0.1 (The rank is at most the smaller dimension)**.** *For any $m \times n$ matrix* $\mathbf{A}$*,* $\operatorname{rank}(\mathbf{A}) \le \min\{m, n\}$.

Proof

Proof. $\mathbf{A}$ has n columns, so at most n of them are linearly independent, and $\operatorname{rank}(\mathbf{A}) \le n$ (Definition 2 in Subspaces and Rank). Likewise $\mathbf{A}^\top$ has m columns, so $\operatorname{rank}(\mathbf{A}) =$

$\operatorname{rank}(\mathbf{A}^\top) \le m$ (Theorem 0.9). □

Example 0.15 (A 2×3 matrix has rank at most 2)**.** By Corollary 0.1, every 2×3 matrix has rank at most $\min\{2, 3\} = 2$, even though it has 3 columns. The matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of Remark 1 in Subspaces and Rank reaches that bound: its first two columns $(1, 0)$ and $(0, 1)$ are linearly independent, so its rank is 2.

Theorem 0.10 (A centered data matrix has rank at most $n-1$)**.** *Let $\mathbf{X}$ be an $n \times p$ matrix with rows $\underset{\sim}{x}_1{}^\top, \dots, \underset{\sim}{x}_n{}^\top$, let $\bar{\underset{\sim}{x}} = \frac{1}{n}\sum_{i=1}^{n} \underset{\sim}{x}_i$ be the mean of the rows, and let $\mathbf{X}_c = \mathbf{X} - \underset{\sim}{1}\,\bar{\underset{\sim}{x}}^\top$ be the matrix whose row i is $(\underset{\sim}{x}_i - \bar{\underset{\sim}{x}})^\top$, where $\underset{\sim}{1}$ is the ones vector of length n (Definition 13 in Vectors). Then* $\operatorname{rank}(\mathbf{X}_c) \le n - 1$.

Proof

Proof. The entries of each column of $\mathbf{X}_c$ sum to zero:

$$\begin{aligned} \underset{\sim}{1}^\top \mathbf{X}_c &= \underset{\sim}{1}^\top \mathbf{X} - (\underset{\sim}{1}^\top \underset{\sim}{1})\bar{\underset{\sim}{x}}^\top && \text{(Theorem 6 in Matrices)} \\ &= \sum_{i=1}^{n} \underset{\sim}{x}_i{}^\top - n\,\bar{\underset{\sim}{x}}^\top && (\underset{\sim}{1}^\top \underset{\sim}{1} = n) \\ &= n\,\bar{\underset{\sim}{x}}^\top - n\,\bar{\underset{\sim}{x}}^\top && \text{(definition of } \bar{\underset{\sim}{x}}) \\ &= \underset{\sim}{0}_p{}^\top. \end{aligned}$$

Transposing, $\mathbf{X}_c{}^\top \underset{\sim}{1} = \underset{\sim}{0}_p$. The ones vector is nonzero, so it is in the null space of the $p \times n$ matrix $\mathbf{X}_c{}^\top$, and $\operatorname{nullity}(\mathbf{X}_c{}^\top) \ge 1$. By Theorem 0.7, applied to the n columns of $\mathbf{X}_c{}^\top$,

$$\begin{aligned} \operatorname{rank}(\mathbf{X}_c) &= \operatorname{rank}(\mathbf{X}_c{}^\top) && \text{(Theorem 0.9)} \\ &= n - \operatorname{nullity}(\mathbf{X}_c{}^\top) && \text{(Theorem 0.7)} \\ &\le n - 1. && (\operatorname{nullity}(\mathbf{X}_c{}^\top) \ge 1) \end{aligned}$$

□

Example 0.16 (Two points in three dimensions)**.** $\mathbf{X} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}$ has $\bar{\underset{\sim}{x}} = (2, 2, 2)$, so

$$\mathbf{X}_c = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & -1 \end{bmatrix}.$$

The second row is -1 times the first, so $\operatorname{rank}(\mathbf{X}_c) = 1 = n - 1$, and the column sums are zero: $\underset{\sim}{1}^\top \mathbf{X}_c = \underset{\sim}{0}_3{}^\top$.

Corollary 0.2 (A sample covariance matrix from $n \leq p$ points is singular)**.** *With* $\mathbf{X}_c$ *as in Theorem 0.10, the sample covariance matrix*

$$\mathbf{S} = \frac{1}{n-1} \mathbf{X}_c{}^\top \mathbf{X}_c$$

is $p \times p$ *with* $\operatorname{rank}(\mathbf{S}) \leq n-1$*. So if* $n \leq p$*, then* $\mathbf{S}$ *is not invertible.*

Proof

Proof.

$$\begin{aligned} \operatorname{rank}(\mathbf{S}) &= \operatorname{rank}(\mathbf{X}_c{}^\top \mathbf{X}_c) && \text{(multiplying by } \tfrac{1}{n-1} \neq 0 \text{ does not change the rank)} \\ &= \operatorname{rank}(\mathbf{X}_c) && \text{(Theorem 0.9)} \\ &\leq n-1. && \text{(Theorem 0.10)} \end{aligned}$$

If $n \leq p$, then $\operatorname{rank}(\mathbf{S}) \leq n-1 < p$, so $\mathbf{S}$ does not have full column rank, and it is not invertible by Theorem 1 in Diagonalization and Cholesky. □

Example 0.17 (As many points as dimensions)**.** The $n = 3$ points $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$ in $p = 3$ dimensions have $\underset{\sim}{\bar{x}} = \frac{1}{3}(1,1,1)$, so

$$\mathbf{X}_c = \mathbf{I}_3 - \tfrac{1}{3} \underset{\sim}{1}\underset{\sim}{1}^\top, \qquad \mathbf{S} = \tfrac{1}{2} \mathbf{X}_c = \tfrac{1}{6} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix},$$

using $\mathbf{X}_c{}^\top \mathbf{X}_c = \mathbf{X}_c$ (the centering matrix is a symmetric projection). Then $\mathbf{S}\underset{\sim}{1} = \underset{\sim}{0}_3$ and $\operatorname{rank}(\mathbf{S}) = 2 = n - 1 < 3$. With one more point than dimensions ($n = p + 1$), the rank bound allows $\mathbf{S}$ to have full rank p.

Definition 0.8 (Affine subspace)**.** An **affine subspace** [5] of $\mathbb{R}^p$ is a set of the form

$$\{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{S}\}$$

for some vector $\underset{\sim}{x}_0 \in \mathbb{R}^p$ and some subspace $\mathcal{S}$ of $\mathbb{R}^p$ (Definition 4 in Subspaces and Rank): the image of $\mathcal{S}$ under the translation by $\underset{\sim}{x}_0$ (Definition 18 in Matrices).

Example 0.18 (A line that misses the origin is an affine subspace)**.** The line $\mathcal{T} = \{(x,1) : x \in \mathbb{R}\}$ of Example 4 in Subspaces and Rank is not a subspace, but it is an affine subspace, with $\underset{\sim}{x}_0 = (0,1)$ and $\mathcal{S} = \operatorname{span}\{(1,0)\}$ (Definition 5 in Subspaces and Rank):

$$
\begin{aligned}
(0,1) + c\,(1,0) &= (0 + c,\ 1 + 0) && \text{(Definition 6 in Matrices, Definition 7 in Vectors)} \\
&= (c, 1), && \text{(add)}
\end{aligned}
$$

and as c ranges over $\mathbb{R}$, $(c, 1)$ ranges over all of $\mathcal{T}$. Every subspace $\mathcal{S}$ is also an affine subspace, with $\underset{\sim}{x}_0 = \underset{\sim}{0}$.

Theorem 0.11 (A hyperplane is a translated subspace of dimension $p-1$)**.** *Let $\underset{\sim}{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, let $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w}^\top \underset{\sim}{x} + b = 0\}$ be the hyperplane with normal vector $\underset{\sim}{w}$ and offset b (Definition 22 in Matrices), and let $\mathcal{H}_0 = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w}^\top \underset{\sim}{x} = 0\}$ be the hyperplane with the same normal vector and offset 0.*

1. *$\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ (Definition 4 in Subspaces and Rank) of dimension $p-1$ (Definition 9 in Subspaces and Rank).*
2. *$\mathcal{H}$ is not empty, and for any $\underset{\sim}{x}_0 \in \mathcal{H}$, $\mathcal{H} = \{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{H}_0\}$, so $\mathcal{H}$ is an affine subspace (Definition 0.8).*
3. *For any $\underset{\sim}{x}, \underset{\sim}{y} \in \mathcal{H}$, the normal vector is orthogonal to their difference: $\underset{\sim}{w} \perp (\underset{\sim}{x} - \underset{\sim}{y})$ (Definition 15 in Vectors).*

Proof

Proof. Here $\underset{\sim}{w}^\top$ is a $1 \times p$ matrix, and $\underset{\sim}{w}^\top \underset{\sim}{x} = \underset{\sim}{w} \cdot \underset{\sim}{x}$ (Example 6 in Matrices).
Part 1. $\mathcal{H}_0$ is the null space $\mathcal{N}(\underset{\sim}{w}^\top)$ (Definition 0.3), so it is a subspace (Theorem 0.2). The p columns of $\underset{\sim}{w}^\top$ are the numbers $w_1, \ldots, w_p$. Some $w_j \neq 0$, because $\underset{\sim}{w} \neq \underset{\sim}{0}$, and that column on its own is linearly independent ($c\,w_j = 0$ forces $c = 0$), so $\operatorname{rank}(\underset{\sim}{w}^\top) \geq 1$ (Definition 2 in Subspaces and Rank); and $\operatorname{rank}(\underset{\sim}{w}^\top) \leq \min\{1, p\} = 1$ (Corollary 0.1). So

$$
\begin{aligned}
\dim(\mathcal{H}_0) &= \operatorname{nullity}(\underset{\sim}{w}^\top) && \text{(Definition 0.7)} \\
&= p - \operatorname{rank}(\underset{\sim}{w}^\top) && \text{(Theorem 0.7, for the } p \text{ columns of } \underset{\sim}{w}^\top) \\
&= p - 1. && \text{(the rank is 1)}
\end{aligned}
$$

Part 2. $\underset{\sim}{w} \cdot \underset{\sim}{w} = w_1^2 + \cdots + w_p^2 > 0$, because some $w_j \neq 0$. So $\underset{\sim}{x}^* \overset{\text{def}}{=} -\frac{b}{\underset{\sim}{w} \cdot \underset{\sim}{w}}\,\underset{\sim}{w}$ is defined, and

$$
\begin{aligned}
\underset{\sim}{w}^\top \underset{\sim}{x}^* + b &= -\frac{b}{\underset{\sim}{w} \cdot \underset{\sim}{w}}\,(\underset{\sim}{w}^\top \underset{\sim}{w}) + b && \text{(homogeneity of } \underset{\sim}{x} \mapsto \underset{\sim}{w}^\top \underset{\sim}{x}\text{, Theorem 10 in Matrices)} \\
&= -b + b && (\underset{\sim}{w}^\top \underset{\sim}{w} = \underset{\sim}{w} \cdot \underset{\sim}{w}) \\
&= 0, && \text{(arithmetic)}
\end{aligned}
$$

so $\underset{\sim}{x}^* \in \mathcal{H}$, and $\mathcal{H}$ is not empty. Now take any $\underset{\sim}{x}_0 \in \mathcal{H}$, so $\underset{\sim}{w}^\top \underset{\sim}{x}_0 = -b$. For any $\underset{\sim}{x} \in \mathbb{R}^p$,

$$\begin{aligned} \underset{\sim}{w}^\top(\underset{\sim}{x}-\underset{\sim}{x}_0) &= \underset{\sim}{w}^\top\underset{\sim}{x}-\underset{\sim}{w}^\top\underset{\sim}{x}_0 && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \underset{\sim}{w}^\top\underset{\sim}{x}-(-b) && (\underset{\sim}{x}_0\in\mathcal{H}) \\ &= \underset{\sim}{w}^\top\underset{\sim}{x}+b, && \text{(arithmetic)} \end{aligned}$$

so $\underset{\sim}{x}\in\mathcal{H}$ exactly when $\underset{\sim}{x}-\underset{\sim}{x}_0\in\mathcal{H}_0$, that is, exactly when $\underset{\sim}{x}=\underset{\sim}{x}_0+\underset{\sim}{s}$ with $\underset{\sim}{s}=\underset{\sim}{x}-\underset{\sim}{x}_0\in\mathcal{H}_0$. $\mathcal{H}_0$ is a subspace (part 1), so $\mathcal{H}$ has the form in Definition 0.8.

Part 3. For $\underset{\sim}{x},\underset{\sim}{y}\in\mathcal{H}$, both $\underset{\sim}{w}^\top\underset{\sim}{x}$ and $\underset{\sim}{w}^\top\underset{\sim}{y}$ equal $-b$, so

$$\begin{aligned} \underset{\sim}{w}\cdot(\underset{\sim}{x}-\underset{\sim}{y}) &= \underset{\sim}{w}^\top\underset{\sim}{x}-\underset{\sim}{w}^\top\underset{\sim}{y} && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= (-b)-(-b) && (\underset{\sim}{x},\underset{\sim}{y}\in\mathcal{H}) \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

□

Example 0.19 (The plane $x_1+x_2+x_3=1$ in $\mathbb{R}^3$)**.** Take $\underset{\sim}{w}=(1,1,1)$ and $b=-1$, so $\mathcal{H}=\{\underset{\sim}{x}:x_1+x_2+x_3-1=0\}$, the plane that Example 5 in Subspaces and Rank showed is not a subspace. By part 1, $\mathcal{H}_0=\{\underset{\sim}{x}:x_1+x_2+x_3=0\}$ is a subspace of dimension $3-1=2$.

- $\underset{\sim}{x}_0=(1,0,0)$ is in $\mathcal{H}$, since $1+0+0-1=0$, and so is $\underset{\sim}{y}=(0,1,0)$.
- Their difference $\underset{\sim}{y}-\underset{\sim}{x}_0=(-1,1,0)$ has entries summing to $-1+1+0=0$, so it is in $\mathcal{H}_0$, and $\underset{\sim}{y}=\underset{\sim}{x}_0+(-1,1,0)$, as part 2 says.
- $$\begin{aligned} \underset{\sim}{w}\cdot(-1,1,0) &= -1+1+0 \\ &= 0, \end{aligned}$$

  as part 3 says.
- The point built in the proof is

  $$\begin{aligned} \underset{\sim}{x}^* &= -\frac{-1}{3}(1,1,1) \\ &= (\tfrac{1}{3},\tfrac{1}{3},\tfrac{1}{3}), \end{aligned}$$

  and $\frac{1}{3}+\frac{1}{3}+\frac{1}{3}-1=0$.

Theorem 0.12 (Where a hyperplane passes, and its shape in $\mathbb{R}^2$ and $\mathbb{R}^3$)**.** *Let* $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0\}$ *be a hyperplane (Definition 22 in Matrices).*

1. $\mathcal{H}$ *contains the origin* $\underset{\sim}{0}$ *exactly when* $b = 0$.
2. *If* $p = 2$*, then* $\mathcal{H}$ *is a line:* $\mathcal{H} = \{\underset{\sim}{x}_0 + t\,\underset{\sim}{v} : t \in \mathbb{R}\}$ *for some* $\underset{\sim}{x}_0 \in \mathbb{R}^2$ *and some nonzero* $\underset{\sim}{v} \in \mathbb{R}^2$.
3. *If* $p = 3$*, then* $\mathcal{H}$ *is a plane:* $\mathcal{H} = \{\underset{\sim}{x}_0 + s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2 : s, t \in \mathbb{R}\}$ *for some* $\underset{\sim}{x}_0 \in \mathbb{R}^3$ *and some linearly independent* $\underset{\sim}{v}_1, \underset{\sim}{v}_2 \in \mathbb{R}^3$.

Proof

Proof. **Part 1.** Since $\underset{\sim}{w} \cdot \underset{\sim}{0} = 0$,

$$\begin{aligned} \underset{\sim}{w} \cdot \underset{\sim}{0} + b &= 0 + b && \text{(the dot product with } \underset{\sim}{0} \text{ is 0)} \\ &= b, && \text{(Theorem 4 in Convexity, Infimum and Sums)} \end{aligned}$$

so $\underset{\sim}{0} \in \mathcal{H}$ exactly when $b = 0$.

Parts 2 and 3. By Theorem 0.11, $\mathcal{H} = \{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{H}_0\}$ for any $\underset{\sim}{x}_0 \in \mathcal{H}$, where $\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ of dimension $p - 1$.

For $p = 2$, $\mathcal{H}_0$ has dimension 1, so it has a basis (Definition 6 in Subspaces and Rank) with one vector $\underset{\sim}{v}$. That vector is nonzero, and $\mathcal{H}_0 = \{t\,\underset{\sim}{v} : t \in \mathbb{R}\}$.

For $p = 3$, $\mathcal{H}_0$ has dimension 2, so it has a basis with two vectors $\underset{\sim}{v}_1$ and $\underset{\sim}{v}_2$. They are linearly independent, and $\mathcal{H}_0 = \{s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2 : s, t \in \mathbb{R}\}$. □

Example 0.20 (A line in $\mathbb{R}^2$ and a plane in $\mathbb{R}^3$, checked on a grid of points)**.** In $\mathbb{R}^2$, take $\underset{\sim}{w} = (1, 1)$ and $b = -1$, the line of Example 20 in Matrices. The point $\underset{\sim}{x}_0 = -\frac{b}{\underset{\sim}{w} \cdot \underset{\sim}{w}}\,\underset{\sim}{w}$ is on it, and the vector $\underset{\sim}{v} = (-w_2, w_1)$ is nonzero with $\underset{\sim}{w} \cdot \underset{\sim}{v} = 0$, so it spans the subspace $\mathcal{H}_0$ of Theorem 0.11. In $\mathbb{R}^3$, take $\underset{\sim}{w} = (1, 1, 1)$ and $b = -1$, the plane of Example 0.19, with the independent vectors $\underset{\sim}{v}_1 = (1, -1, 0)$ and $\underset{\sim}{v}_2 = (0, 1, -1)$ in $\mathcal{H}_0$. The code evaluates $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ at the origin and at points $\underset{\sim}{x}_0 + t\,\underset{\sim}{v}$ and $\underset{\sim}{x}_0 + s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2$.

```r
level <- function(w, b, x) sum(w * x) + b

w2 <- c(1, 1)
b2 <- -1
point_line <- -b2 / sum(w2 * w2) * w2
v <- c(-w2[2], w2[1])
line_levels <- sapply(-2:2, function(t) level(w2, b2, point_line + t * v))

w3 <- c(1, 1, 1)
b3 <- -1
point_plane <- -b3 / sum(w3 * w3) * w3
v1 <- c(1, -1, 0)
v2 <- c(0, 1, -1)
grid <- expand.grid(s = -2:2, t = -2:2)
plane_levels <- mapply(
  function(s, t) level(w3, b3, point_plane + s * v1 + t * v2),
  grid$s, grid$t
)

c(
  origin_in_R2 = level(w2, b2, c(0, 0)),
  origin_in_R3 = level(w3, b3, c(0, 0, 0)),
  origin_with_b0 = level(w3, 0, c(0, 0, 0))
)
#>   origin_in_R2   origin_in_R3 origin_with_b0
#>             -1             -1              0
```

At the origin the left side equals the offset, so the origin is on the line only if b is 0: it gives -1 for the line and -1 for the plane, and 0 once the offset is changed to 0. All 5 points $\underset{\sim}{x}_0 + t\,\underset{\sim}{v}$ on the line and all 25 points $\underset{\sim}{x}_0 + s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2$ on the plane satisfy the equation up to rounding error: 30 of the 30 points have $|\underset{\sim}{w} \cdot \underset{\sim}{x} + b| < 10^{-10}$.

Definition 0.9 (Open half-space)**.** Let $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0\}$ be a hyperplane (Definition 22 in Matrices). The two **open half-spaces** [6] of $\mathcal{H}$ are

$$\mathcal{H}^+ \stackrel{\text{def}}{=} \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b > 0\} \quad \text{and} \quad \mathcal{H}^- \stackrel{\text{def}}{=} \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b < 0\}. \tag{1}$$

Definition 0.10 (Closed half-space)**.** The **closed half-spaces** of the hyperplane $\mathcal{H}$ of Definition 0.9 also include $\mathcal{H}$ itself: $\mathcal{H}^+ \cup \mathcal{H}$ and $\mathcal{H}^- \cup \mathcal{H}$.

Remark

Remark. Replacing $(\underset{\sim}{w}, b)$ by $(-\underset{\sim}{w}, -b)$ describes the same hyperplane but swaps the names $\mathcal{H}^+$ and $\mathcal{H}^-$. The sign of $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ tells you which side of the hyperplane $\underset{\sim}{x}$ is on. A linear classifier uses this sign to assign one of two labels.

Example 0.21 (Half-spaces of a line in the plane)**.** Let $\mathcal{H}$ be the line in $\mathbb{R}^2$ with $\underset{\sim}{w} = (1, 2)$ and $b = -2$ (Definition 22 in Matrices), so $\mathcal{H} = \{\underset{\sim}{x} : x_1 + 2x_2 - 2 = 0\}$. Evaluate $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ at three points (Definition 0.9):

- At $(0, 0)$ it equals $0 + 0 - 2 = -2 < 0$, so $(0, 0) \in \mathcal{H}^-$.
- At $(2, 0)$ it equals $2 + 0 - 2 = 0$, so $(2, 0) \in \mathcal{H}$.
- At $(2, 1)$ it equals $2 + 2 - 2 = 2 > 0$, so $(2, 1) \in \mathcal{H}^+$.

Moving from $(2, 0)$ in the direction $\underset{\sim}{w} = (1, 2)$ gives $(3, 2)$, where the value is $3 + 4 - 2 = 5 > 0$. So the direction $\underset{\sim}{w}$ leads into $\mathcal{H}^+$.

Remark (The two sides of a hyperplane)

Remark 0.1 (The two sides of a hyperplane). Every point of $\mathbb{R}^p$ is in exactly one of $\mathcal{H}^+$, $\mathcal{H}$ and $\mathcal{H}^-$, because the number $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ is positive, zero or negative. The vector $\underset{\sim}{w}$ points into $\mathcal{H}^+$: the number $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ goes up when $\underset{\sim}{x}$ moves in the direction $\underset{\sim}{w}$.

Theorem 0.13 (Differences of points in a hyperplane and in a half-space)**.** *Let $\underset{\sim}{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, and let $\mathcal{H}$, $\mathcal{H}^+$ be the hyperplane and open half-space of Definition 0.9. For a set $S \subseteq \mathbb{R}^p$, let $D(S) \overset{def}{=} \{\underset{\sim}{y} - \underset{\sim}{x} : \underset{\sim}{x}, \underset{\sim}{y} \in S\}$ be the set of differences of its points.*

1. *$D(\mathcal{H}) = \{\underset{\sim}{s} : \underset{\sim}{w} \cdot \underset{\sim}{s} = 0\}$, a subspace of dimension $p - 1$.*
2. *$D(\mathcal{H}^+) = \mathbb{R}^p$, a subspace of dimension p.*

The same holds for $\mathcal{H}^-$.

Proof

Proof. **Part 1.** Let $\underset{\sim}{x}, \underset{\sim}{y} \in \mathcal{H}$. By Theorem 0.11 part 3, $\underset{\sim}{w} \cdot (\underset{\sim}{y} - \underset{\sim}{x}) = 0$, so $\underset{\sim}{y} - \underset{\sim}{x} \in \mathcal{H}_0 = \{\underset{\sim}{s} : \underset{\sim}{w} \cdot \underset{\sim}{s} = 0\}$. Conversely, take any $\underset{\sim}{x_0} \in \mathcal{H}$ (Theorem 0.11 part 2 gives such a point). For any $\underset{\sim}{s} \in \mathcal{H}_0$, the point $\underset{\sim}{x_0} + \underset{\sim}{s}$ is in $\mathcal{H}$ by the same part, and $(\underset{\sim}{x_0} + \underset{\sim}{s}) - \underset{\sim}{x_0} = \underset{\sim}{s}$. So $D(\mathcal{H}) = \mathcal{H}_0$, which has dimension $p - 1$ by Theorem 0.11 part 1.
Part 2. Every difference is in $\mathbb{R}^p$, so $D(\mathcal{H}^+) \subseteq \mathbb{R}^p$. For the reverse inclusion, take any $\underset{\sim}{x_0} \in \mathcal{H}$. Take any $\underset{\sim}{v} \in \mathbb{R}^p$. For a number $t > 0$ to be chosen, let $\underset{\sim}{x_t} \overset{\text{def}}{=} \underset{\sim}{x_0} + t\,\underset{\sim}{w}$. Then

$$\begin{aligned} \underset{\sim}{w} \cdot \underset{\sim}{x}_t + b &= (\underset{\sim}{w} \cdot \underset{\sim}{x}_0 + b) + t\, \underset{\sim}{w} \cdot \underset{\sim}{w} && \text{(linearity of } \underset{\sim}{x} \mapsto \underset{\sim}{w} \cdot \underset{\sim}{x}) \\ &= t\, \underset{\sim}{w} \cdot \underset{\sim}{w}, && (\underset{\sim}{x}_0 \in \mathcal{H}) \end{aligned}$$

which is positive, so $\underset{\sim}{x}_t \in \mathcal{H}^+$. Next,

$$\begin{aligned} \underset{\sim}{w} \cdot (\underset{\sim}{x}_t + \underset{\sim}{v}) + b &= (\underset{\sim}{w} \cdot \underset{\sim}{x}_t + b) + \underset{\sim}{w} \cdot \underset{\sim}{v} && \text{(linearity of } \underset{\sim}{x} \mapsto \underset{\sim}{w} \cdot \underset{\sim}{x}) \\ &= t\, \underset{\sim}{w} \cdot \underset{\sim}{w} + \underset{\sim}{w} \cdot \underset{\sim}{v}. && \text{(the display above)} \end{aligned}$$

Because $\underset{\sim}{w} \cdot \underset{\sim}{w} > 0$, this is positive once $t > -\underset{\sim}{w} \cdot \underset{\sim}{v} / \underset{\sim}{w} \cdot \underset{\sim}{w}$. Choose such a $t > 0$. Then $\underset{\sim}{x}_t$ and $\underset{\sim}{x}_t + \underset{\sim}{v}$ are both in $\mathcal{H}^+$, and their difference is $\underset{\sim}{v}$. So $\underset{\sim}{v} \in D(\mathcal{H}^+)$. The whole space $\mathbb{R}^p$ is a subspace of dimension p (Definition 9 in Subspaces and Rank).
The proof for $\mathcal{H}^-$ is the same with $-\underset{\sim}{w}$ and $-b$ in place of $\underset{\sim}{w}$ and b. □

Example 0.22 (Two iris species on either side of a line)**.** The `iris` data set in R has the petal length and petal width, in centimeters, of 150 irises. Take the 100 flowers of the species versicolor and virginica, and let $\underset{\sim}{x} = (x_1, x_2)$ be the petal length and petal width of one flower. A logistic regression of the species on $\underset{\sim}{x}$ gives a coefficient vector $\underset{\sim}{w}$ and an intercept b. The line $\underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0$ is a hyperplane in $\mathbb{R}^2$ (Definition 22 in Matrices), and the two species mostly fall in its two open half-spaces (Definition 0.9). Figure 1 shows the line and the flowers.

```
two_species <- dplyr::filter(iris, Species != "setosa") |> droplevels()
fit <- glm(
  Species ~ Petal.Length + Petal.Width,
  family = binomial,
  data = two_species
)
w <- coef(fit)[c("Petal.Length", "Petal.Width")]
b <- coef(fit)[["(Intercept)"]]
two_species$score <- as.vector(
  as.matrix(two_species[c("Petal.Length", "Petal.Width")]) %*% w + b
)
two_species$side <- ifelse(two_species$score > 0, "positive", "negative")
ggplot2::ggplot(two_species) +
  ggplot2::aes(x = Petal.Length, y = Petal.Width, colour = Species) +
  ggplot2::geom_point() +
  ggplot2::geom_abline(intercept = -b / w[[2]], slope = -w[[1]] / w[[2]]) +
  ggplot2::labs(x = "Petal length (cm)", y = "Petal width (cm)")
```

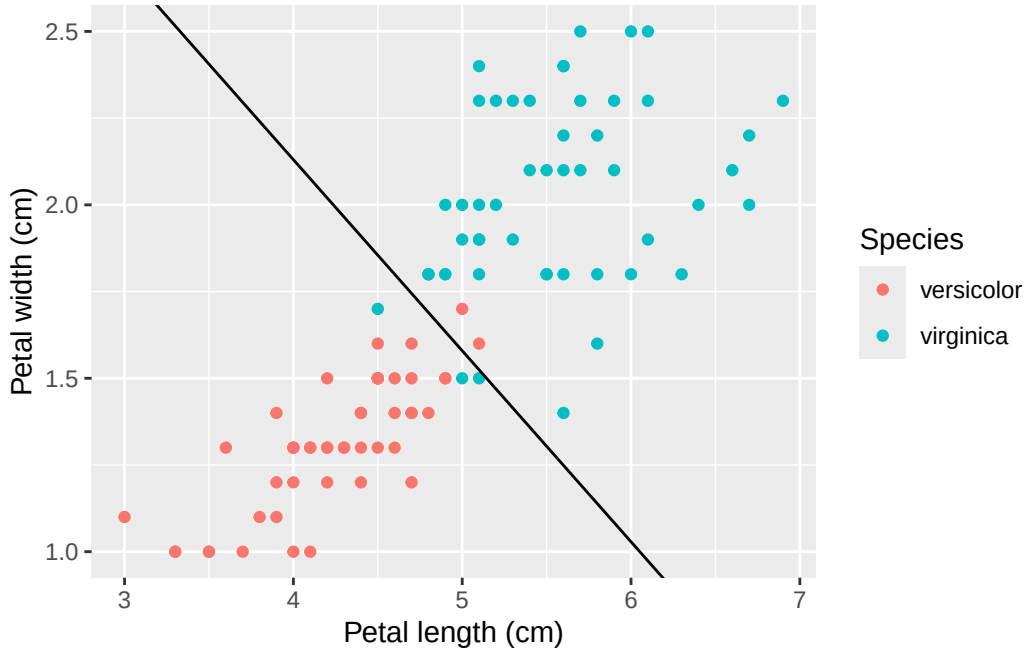


Figure 1: Petal length and width of versicolor and virginica irises, with the line where a logistic regression is indifferent between the two species

```
table(two_species$Species, two_species$side)
#>
#>              negative positive
#>   versicolor       47        3
#>   virginica         3       47
```

The vector $\underset{\sim}{w}$ points toward the virginica side, so the positive half-space $\mathcal{H}^+$ holds 50 of the 100 flowers, and 47 of them are virginica. The line is a set of dimension $p-1=1$, and each side is a region of the plane of dimension $p=2$ (Theorem 0.13).

Exercise 0.1 (Which side of the plane?)**.** Let $\mathcal{H}$ be the plane in $\mathbb{R}^3$ with $\underset{\sim}{w}=(1,1,1)$ and $b=-1$ (Example 0.19).

1. For each point, say whether it is in $\mathcal{H}^+$, $\mathcal{H}$ or $\mathcal{H}^-$ (Definition 0.9): $(0,0,0)$, $(1,0,0)$ and $(1,1,1)$.
2. Write the plane and its half-spaces using $-\underset{\sim}{w}$ and $-b$. Which points from part 1 change sides?
3. State the dimension of $D(\mathcal{H})$ and of $D(\mathcal{H}^+)$ (Theorem 0.13).

Solution

Solution 0.1.

1. Compute $\underset{\sim}{w} \cdot \underset{\sim}{x} + b = x_1 + x_2 + x_3 - 1$ at each point.
   - At $(0,0,0)$ it equals $0+0+0-1 = -1 < 0$, so the point is in $\mathcal{H}^-$.
   - At $(1,0,0)$ it equals $1+0+0-1 = 0$, so the point is in $\mathcal{H}$.
   - At $(1,1,1)$ it equals $1+1+1-1 = 2 > 0$, so the point is in $\mathcal{H}^+$.
2. With $-\underset{\sim}{w} = (-1,-1,-1)$ and $-b = 1$, the equation is $-x_1 - x_2 - x_3 + 1 = 0$. Multiplying both sides by -1 gives the original equation, so it is the same plane. The new expression is the negative of the old one at every point, so the new positive half-space is the old $\mathcal{H}^-$ and the new negative half-space is the old $\mathcal{H}^+$. The origin $(0,0,0)$ is now in the positive half-space, and $(1,1,1)$ is now in the negative half-space. The point $(1,0,0)$ stays on the plane. So the origin and $(1,1,1)$ both change names, and the point on the plane does not.
3. Here $p = 3$. $D(\mathcal{H})$ is the subspace $\{\underset{\sim}{s} : s_1 + s_2 + s_3 = 0\}$, which has dimension $p - 1 = 2$. $D(\mathcal{H}^+) = \mathbb{R}^3$ has dimension $p = 3$.

Definition 0.11 (Rank factorization)**.** Let $\mathbf{A}$ be an $m \times n$ matrix with $\operatorname{rank}(\mathbf{A}) = r \geq 1$. A **rank factorization** [7] of $\mathbf{A}$ is a product

$$\underset{m\times n}{\mathbf{A}} = \underset{m\times r}{\mathbf{C}}\ \underset{r\times n}{\mathbf{R}}\ .$$

Example 0.23 (A rank factorization of a rank-one matrix)**.** The matrix $\mathbf{A}$ in Example 0.1 has rank 1 (Example 0.10), and

$$\begin{bmatrix}1\\3\end{bmatrix}\begin{bmatrix}1 & -2 & -2\end{bmatrix} = \begin{bmatrix}1\cdot 1 & 1\cdot(-2) & 1\cdot(-2)\\ 3\cdot 1 & 3\cdot(-2) & 3\cdot(-2)\end{bmatrix} = \begin{bmatrix}1 & -2 & -2\\ 3 & -6 & -6\end{bmatrix},$$

so $\mathbf{C} = \begin{bmatrix}1\\3\end{bmatrix}$ (2×1) and $\mathbf{R} = \begin{bmatrix}1 & -2 & -2\end{bmatrix}$ (1×3) are a rank factorization. It is not unique: $\mathbf{C} = \begin{bmatrix}2\\6\end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix}\frac{1}{2} & -1 & -1\end{bmatrix}$ give the same product. A product $\mathbf{CR}$ with $\mathbf{C}$ of size 2×2 is not a rank factorization of this $\mathbf{A}$, because the inner dimension must equal $\operatorname{rank}(\mathbf{A}) = 1$.

Theorem 0.14 (Every nonzero matrix has a rank factorization)**.** *Every $m \times n$ matrix $\mathbf{A}$ with $\operatorname{rank}(\mathbf{A}) = r \geq 1$ has a rank factorization (Definition 0.11). One is given by taking the columns of $\mathbf{C}$ to be any basis of $\mathcal{C}(\mathbf{A})$.*

Proof

Proof. $\mathcal{C}(\mathbf{A})$ has dimension r (Theorem 0.6), so it has a basis $\underset{\sim}{c}_1, \ldots, \underset{\sim}{c}_r$ (Theorem 9 in Subspaces and Rank); let $\mathbf{C}$ be the $m \times r$ matrix with these columns. Each column $\underset{\sim}{a}_j$ of $\mathbf{A}$ is in $\mathcal{C}(\mathbf{A})$, so $\underset{\sim}{a}_j = r_{1j}\underset{\sim}{c}_1 + \cdots + r_{rj}\underset{\sim}{c}_r$ for some numbers r_{ij}, because the basis spans $\mathcal{C}(\mathbf{A})$. Let $\mathbf{R}$ be the $r \times n$ matrix with entries r_{ij}, whose column j is $\underset{\sim}{r}_j = (r_{1j}, \ldots, r_{rj})$. Then column j of $\mathbf{CR}$ is

$$\begin{aligned} \mathbf{C}\underset{\sim}{r}_j &= r_{1j}\underset{\sim}{c}_1 + \cdots + r_{rj}\underset{\sim}{c}_r && \text{(Theorem 3 in Subspaces and Rank)} \\ &= \underset{\sim}{a}_j, && \text{(choice of the } r_{ij}\text{)} \end{aligned}$$

so $\mathbf{CR} = \mathbf{A}$. □

Example 0.24 (Building a rank factorization from a basis)**.** For $\mathbf{A}$ in Example 0.1, $(1, 3)$ spans $\mathcal{C}(\mathbf{A})$ (Example 0.2), and it is linearly independent because it is nonzero ($c\,(1, 3) = \underset{\sim}{0}_2$ forces $c = 0$), so it is a basis of $\mathcal{C}(\mathbf{A})$. The columns of $\mathbf{A}$ are $(1, 3) = 1 \cdot (1, 3)$, $(-2, -6) = -2 \cdot (1, 3)$ and $(-2, -6) = -2 \cdot (1, 3)$, so the construction in Theorem 0.14 gives $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$, the factorization in Example 0.23.

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